

SEAN LIN

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🌐 [LinkedIn](#)

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Education

Stanford University

Sep. 2023 – Dec. 2026

M.S. in Aeronautical and Astronautical Engineering (3.86/4.00)

Stanford, CA

- Relevant Coursework: Classical Dynamics, Orbital Mechanics, Advanced Feedback Control Design, Optimal Control
- Note: Took a Leave-of-Absence from Jan 2024 to April 2025

University of California, Berkeley

Aug. 2019 – May 2023

B.S. in Mechanical Engineering, B.A. in Business Administration, Minor in EECS (3.93/4.00)

Berkeley, CA

- M.E.T. Program; High Honors; Regents and Chancellor's Scholar; ASME; Tau Beta Pi

Experience

HyLight

September 2024 – March 2025

Airship Dynamics Engineering Intern

Paris, FR

- Designed a bench that reads thrust, torque, vibration, RPM, and power data from novel electric-propulsion-systems.
- Characterized aerodynamic coefficients of airship across attitudes and airspeeds for flight controller implementation.
- Reverse-engineered a ballonet system for buoyancy control. Sensor input from altimeter and temperature sensor.

De-Ice

June 2024 – Aug 2024

Thermals and Dynamics Intern

Somerville, MA

- Cradle-to-the-grave design and assembly of a bench test setup to mimic in-flight avionics thermal conditions. Applied first principles of fluids and heat transfer for analytical validation.
- Perform vibration analysis on avionics. Propose solutions for 1st or 2nd order resonant coupling problems.

NASA Ames Research Center

September 2024 – May 2025

Mechanical Engineer (Part Time)

Mountain View, CA

- Digitized 100+ mechanical drawings for the Electric Arc Shock Tube Facility (EAST).
- Perform structural analysis on high-pressure parts prior to design/material changes.

SpaceX

June 2023 – September 2023

Associate Avionics Engineer

Hawthorne, CA

- Design a Destructive Physical Analysis routine on avionics EEE parts. Findings lead to a containment of \$45000.
- Perform electrical Failure Analysis (FA) of pressure transducers and capacitors.
- Validated Raptor engine MEMS and piezoelectric oscillators under helium-rich environments using Physics-of-Failure.

SpaceX

May 2022 – August 2022

Launch Engineering Intern

Cape Canaveral, FL

- Full design ownership of 4 Ground-Station-Equipment and launch pad hardware assemblies.
- Closed any open supply chain technical issues for all sleeve bearings used on the launch pad.

Research

Institute for Flight Mechanics and Control

Jun 2026 – Present

Visiting Researcher - PI: Cunis Torbjorn

Stuttgart, Germany

- Controllability-based design of VLEO satellites in rarified atmospheric orbit.
- Designed an optimization workflow for connecting control-surface configuration with desired vehicle torque.
- Defined key torque gradients to implement in the controller.

Space Rendez-Vous Lab

Jan 2026 – Present

Student Researcher - PI: Simone D'Amico

Stanford, CA

- Design a vision-based satellite navigation workflow for improved space-situational-awareness (SSA).
- Propagate orbits of a catalog of 20,000+ satellites and perform a detectability analysis on visible satellites.
- Designed a catalog-refining feedback workflow where observer and target satellites reinforce each-others positions.

Skills

Software and Embedded Systems: Python, MATLAB, C++, Java, Julia, SQL, ESP32, Arduino, Raspberry Pi

Engineering: SolidWorks, ANSYS, NX, AutoCAD, Onshape, EasyEDA, SimScale, MATLAB, LTSpice

Languages: English (L1 native), Mandarin (C1), French (B2)

Sean Lin

CONTACT INFORMATION	216 Rosse Ave Stanford, CA 94305	(909) 538-7519 seanlin@stanford.edu https://slin1923-portfolio.github.io/
ACADEMIC INTERESTS	Spacecraft Guidance, Navigation, and Control; ADCS; Space-Situational-Awareness; Vehicle Autonomy; Control Theory	
EDUCATION	Stanford University, Stanford, CA M.S. Aeronautics and Astronautics, Expected December 2026 - GPA: 3.86/4.00 University of California Berkeley, Berkeley, CA B.S. Mechanical Engineering, May 2023 B.S. Business Administration (Haas), May 2023 Minor Electrical Engineering, May 2023 - High Honors - GPA: 3.93/4.00	
RESEARCH EXPERIENCE	iFR @ University of Stuttgart P.I. Cunis Torbjørn Jun 2026 - Present - Controllability-based design of VLEO satellites in rarified atmospheric orbit. - Designed an optimization workflow for connecting control-surface configuration with desired vehicle torque. - Defined key torque gradients to implement in the controller. Space Rendez-Vous Lab @ Stanford P.I. Simone D'Amico Jan 2026 - Present - Design a vision-based satellite navigation workflow for improved space-situational-awareness (SSA). - Propagate orbits of a catalog of 20,000+ satellites and perform a detectability analysis on visible satellites. - Designed a catalog-refining feedback workflow where observer and target satellites reinforce each-others positions. FLOW Lab @ Berkeley P.I. Simo Makiharju Sept 2021 - May 2022 - Design and assembly of Magnetohydrodynamic Pump as a demonstration of Lorentz-Force pressure gradients. The Kulik Group @ MIT P.I. Heather Kulik June 2018 - Sept 2018 - Visiting student computational chemistry researcher.	
TEACHING EXPERIENCE	ME 106: Fluid Mechanics Course Staff Aug 2022 - Dec 2022 - Design and grade PSET and exam rubrics - Lead weekly online discussion and section	

INDUSTRY
EXPERIENCE

EECS 16B: Designing Information Devices and Systems II

Course Staff

Aug 2021 - May 2022

- Design and grade PSET and exam rubrics

HyLight

Propulsion and Dynamics Intern

Sep 2024 - Mar 2025

Le Plessis-Pate, FR

- Designed an electric propulsion system test bench with Torque, Thrust, Power, and Vibration readings.
- CFD aerodynamic analysis.
- Designed a buoyancy-regulator to replace off-the-shelf seals and valves.

De-Ice

Thermal Engineering Intern

Jun 2024 - Sep 2024

Boston, MA

- Designed a testing apparatus to mimic in-flight thermal conditions for full-power UHF test monitoring.
- Perform shock and vibration analysis on avionics and propose solutions to low order resonant coupling.

NASA Ames Research Center

Mechanical Design Intern

Sep 2024 - Mar 2025

Mountain View, CA

- Design and integrate legacy shock tube parts for Electric Arc Shock Tube.
- Perform high-pressure structural analysis on critical parts.

SpaceX

Associate Avionics Engineer

Jun 2023 - Sep 2023

Hawthorne, CA

- Design a Destructive Physical Analysis lab routine on avionics EEE parts. Findings lead to a faulty part containment of \$45,000.
- Perform electrical Failure Analysis on flight pressure transducers and capacitors.
- Validated Raptor engine oscillators under helium environments.

Launch Engineering Intern

May 2022 - Aug 2022

Cape Canaveral, FL

- Owns 4 assemblies for Ground-Station-Equipment and launch pad hardware.
- Closed open supply chain issues for all sleeve bearings used on the launch pad.

FormFactor Inc.

Engineering Intern

May 2021 - Aug 2021

Livermore, CA

- Silicon wafer metrology and error analysis.

Hermes Robotics

Engineering Intern

May 2020 - Aug 2020

Berkeley, CA

- Mechanical design of an actuator system that autonomously shift's a gearstick.

GRANTS & AWARDS	University of California, Berkeley Regents and Chancellors Scholarship California Alumni Scholarship	Aug 2019 - May 2023 Aug 2019 - May 2023
PUBLICATIONS	HJ. Kulik et. al., <i>Designing in the face of uncertainty: exploiting electronic structure and machine learning models for discovery in inorganic chemistry</i> , Inorganic Chemistry, Cambridge, MA, 2019. HJ. Kulik et. al., <i>Seeing is believing: Experimental spin states from machine learning model structure predictions</i> , The Journal of Physical Chemistry, Cambridge, MA, 2020.	
LANGUAGES	English (Native), Mandarin Chinese (C1), French (B2), German (A1)	
PROGRAMMING LANGUAGES	Python, C++, Matlab, Julia, Java, bash, L ^A T _E X, ROS2,	
REFERENCES	D’Amico, Simone , Professor, Stanford University, damicos@stanford.edu Cunis, Torbjorn , Lecturer, iFR Stuttgart, torbjoern.cunis@ifr.uni-stuttgart.de Jaime, Herzberger , EEE Manager, SpaceX, Jaemi.Herzberger@spacex.com Makiharju, Simo , Professor, UC Berkeley, makiharju@berkeley.edu Trejo, Layla , Cape Launch Hardware, SpaceX, Layla.Trejo@spacex.com	